

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2023

Course: B. Tech. Branch :Mechanical Engineering Semester : VI

Subject Code & Name: BTMPE603A - IC Engines

Max Marks: 60

Date: 17 -07- 2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|----|--|--------------|----------|
| A) | Explain with neat diagram working of Two Stroke Spark Ignition engine. | (Understand) | 6 |
| B) | Explain the losses involved in actual fuel-air cycle. | (Understand) | 6 |
| C) | Describe the following Engine Nomenclature: | (Understand) | 6 |
| | 1)Swept Volume. | | |
| | 2)Clearance Volume. | | |
| | 3)Compression Ratio. | | |
| | (Draw a neat sketch to define above terms). | | |

Q.2 Solve Any Two of the following.

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|----|--|--------------|----------|
| A) | Explain with the help of P-θ diagram, different stages of combustion in Compression Ignition engine. | (Understand) | 6 |
| B) | Describe the phenomenon of Detonation in Spark Ignition Engine. | (Understand) | 6 |
| C) | What are the objectives of combustion chamber design for CI Engine? | (Understand) | 6 |

Q. 3 Solve Any Two of the following.

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|----|--|--------------|----------|
| A) | Explain the working of Simple Carburetor with neat diagram and states its limitations. | (Understand) | 6 |
| B) | Explain Battery Ignition System with neat diagram | (Understand) | 6 |
| C) | Discuss “Force circulation cooling system” of engine. | (Understand) | 6 |

Q.4 Solve Any Two of the following.

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|----|--|---------------|----------|
| A) | What do you mean by super charger? Discuss its effect on – (i)Power output, (ii) Mechanical efficiency (iii) Fuel consumption | (Understand) | 6 |
| B) | What are the ways to control engine emissions? Explain any one in details | (Understand) | 6 |
| C) | During trial of a single cylinder, 4 stroke oil engine the following results were obtained:
Cyl bore=200mm, Stroke=400mm, mep=6 bar, Torque=407Nm, speed=250 RPM, Oil consumption=4kg/hr, CV of fuel=43MJ/kg, | (Application) | 6 |

Cooling water rate=4.5kg/min, Air used per kg of fuel= 30kg, Rise in cooling water temp=45°C, Temp of Exhaust gases=420°C, Room temp=20°C, mean sp. heat of exhaust gases=1kJ/kgK, Sp. Heat of water=4.18kJ/kgK, Barometric pressure=1.01325 bar Find IP, BP and draw up heat balance sheet in kJ/hr.

Q. 5 Solve Any Two of the following.

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|----|---|--------------|----------|
| A) | Explain Fuel Cell Technology with a neat diagram. | (Understand) | 6 |
| B) | Why there is need of alternative fuels? List out different alternative fuels of engine and explain any one. | (Understand) | 6 |
| C) | Compare Electric vehicle with Hybrid Electric vehicle and list advantages of Hybrid vehicle | (Understand) | 6 |

***** End *****